

PHOENIX – Risk Score & Threat Classification Mapping

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Project:	PHOENIX
Deliverable:	Threat Analysis Integration - Risk Score and Threat Classification Mapping

1. Purpose

The purpose of this document is to define the exact mapping logic and decision matrix for the PHOENIX Threat Analysis Integration. It establishes how rule-based threat detections and ADCRS AI risk scoring translate into standard severity ratings, confidence levels, and automated response actions across the system.

This mapping ensures that when an event is processed by the Threat Analysis Engine and ADCRS, the Response Decision Manager can deterministically assign the appropriate mitigation action, dashboard status, and notification trigger.

2. Traceability & Integration Alignment

This deliverable completes the analytical logic required by our existing team artifacts:

[Source Alignment: Mannat & Dilano – Design Doc Sec. 3 & Sequence Diagram Sec. 3 (Step 5 & 7)]

- Threat Scope Alignment:** Directly covers all 9 threat categories defined in Mannat's Threat Analysis Integration Design Document.
- Engine Workflow:** Operationalises the scoring step (ADCRS Risk Scoring Engine) and decision step (Response Decision Manager) documented in Dilano's Integration Workflow and Sequence Diagrams.
- Testing Plan Verification:** Provides the expected values for testing cases FIT-04 (Threat Identification), FIT-05 (Risk Scoring), FIT-07 (Classification), and FIT-08 (Response Decision Processing).

3. Risk Scoring & Severity Thresholds

ADCRS outputs a normalized risk score from 0 to 100 alongside a confidence score between 0.00 and 1.00. The table below defines how score ranges map to severity levels and system handling postures:

Risk Score Range	Severity Level	Confidence Threshold	Response Posture	Notification Trigger
0 – 29	Low	>= 0.50	Informational logging only; normal persistence.	No
30 – 59	Medium	>= 0.65	Flagged on dashboard; standard telemetry tracking.	No / Optional
60 – 84	High	>= 0.75	Targeted mitigation	Yes (Security Ops)

			triggered; security alert generated.	
85 – 100	Critical	>= 0.80	Immediate automated containment; high-priority alert.	Yes (Immediate Broadcast)

4. Comprehensive Threat Classification & Mitigation Matrix

[Source Alignment: Mannat & Dilano – Design Doc Sec. 3 (9 Threat Categories) & Sequence Diagram Sec. 5]

The table below maps all 9 scoped threat categories to their detection source, default severity band, ADCRS score threshold, recommended mitigation action, and output status:

Threat Category	Detection Source	Severity	ADCRS Risk Range	Recommended Action	Status
Login Attacks	TEAVS / Auth Logs	High	75 – 90	lock_account / enforce_mfa	detected
API Abuse	API Gateway Logs	Medium / High	60 – 85	rate_limit_ip / revoke_token	detected
Data Breaches	System Logs / DB	Critical	85 – 100	isolate_database / revoke_keys	detected
Phishing Attacks	App Logs / Email	Medium	45 – 70	quarantine_email / alert_user	detected
Malware & Ransomware	System / Endpoint	Critical	90 – 100	isolate_host / kill_process	detected
DDoS Attacks	API / Network Logs	High / Critical	80 – 95	enable_ddos_shield / block_ip_range	detected
IoT Attacks	IoT Device Events	High	70 – 88	quarantine_device / block_mac	detected
Emergency Alert Compromise	TEAVS Alerts	Critical	90 – 100	suspend_alert_channel / trigger_override	detected
Fraudulent Claims	App Events / Logs	Medium	50 – 75	flag_claim_review / hold_payout	detected

5. Response Decision Logic & Confidence Weighting

The Response Decision Manager calculates final mitigation instructions based on two input vectors: Rule Severity from the Threat Analysis Engine and Risk Score / Confidence from ADCRS.

- **High Risk + High Confidence (Score >= 75, Conf >= 0.75):** Execute automated containment action immediately (e.g., lock_account, isolate_host) and trigger notification.

- **High Risk + Low Confidence (Score ≥ 75 , Conf < 0.75):** Downgrade automated block to 'flag_for_analyst_review' to prevent false positive disruptions while notifying Security Ops.
- **Low Risk + High Confidence (Score < 40 , Conf ≥ 0.80):** Mark status as 'dismissed_clean' or 'logged_info' without triggering alerts.

6. Summary

This Risk Score and Threat Classification Mapping completes the decision logic layer of the PHOENIX Threat Analysis Integration. By defining unambiguous scoring bands, confidence checks, and actionable mitigation codes for all 9 threat categories, it guarantees that automated containment occurs reliably, false positives are minimized, and security analysts receive clear, actionable data.